MONTELUKAST

Neuropsychiatric warning

Evening dose

CHEMICAL NAME [1-({[(1R)-1-{3-[(E)-2-(7-chloroquinolin-2-yl)ethenyl]phenyl}-3-[2-(2-hydroxypropan-2-yl)phenyl]propyl]sulfanyl}methyl)cyclopropyl]acetic acid (sodium)

CLASS Leukotriene receptor antagonist (LTRA)

TRADE NAMES Singulair · Montair · Airlukast · Monkasta

THERAPEUTIC USES / INDICATIONS

- Asthma — add-on to ICS; exercise-induced bronchoconstriction; aspirin-exacerbated asthma.
- Allergic rhinitis (when other treatments unsuitable).
- Preschool viral-induced wheeze (selected children).

ADMINISTRATION

- **PO** once daily in the evening, with or without food.
- Chewable tablets for children; granules can be mixed with soft food (use within 15 min).

DOSAGES

- **Adult & ≥15 yr:** 10 mg once daily.
- **6–14 yr:** 5 mg chewable once daily.
- **6 months–5 yr:** 4 mg granules/chewable once daily.
- **Exercise prevention:** dose ≥2 h before exercise (if not on daily dose).

PHARMACOKINETICS

- **Onset:** effect within 1 day. **Peak:** 3–4 h.
- Metabolised in liver (CYP2C8, 3A4, 2C9).
- Excreted in bile. **t½:** 3–6 h.

PHARMACODYNAMICS

- Blocks **cysteinyl leukotriene (CysLT1) receptors** → ↓ bronchoconstriction, mucus, eosinophilic inflammation and vascular permeability.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, abdominal pain.
- Upper respiratory infection, fever.
- Thirst (children).

ADVERSE EFFECTS (SERIOUS)

- Neuropsychiatric events: sleep disturbance, nightmares, agitation, aggression, depression, suicidal thoughts (boxed warning).
- Churg–Strauss syndrome (EGPA) when steroids reduced (rare).
- Hepatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** history of psychiatric disorder; phenylketonuria (chewables contain aspartame).

KEY INTERACTIONS

- **Phenobarbital, rifampicin:** ↓ levels.
- **Gemfibrozil:** ↑ levels.

NURSING MANAGEMENT

- Ask parents about mood/behaviour/sleep changes.
- Assess asthma control; continue ICS.
- Not for acute attacks.

PATIENT EDUCATION

- Take every evening; not a reliever.
- Report nightmares, mood changes, aggression or thoughts of self-harm immediately.
- Keep using your steroid inhaler.

KEY POINTS

- Leukotriene receptor antagonist — add-on.
- Neuropsychiatric effects; warn families.
- 10 mg evening (adults).
- Not for acute relief.

SOURCES CHECKED **[473]** Montelukast 10 mg Film Coated Tablets (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158) **[474]** Montelukast, oral (US label) **[475]** Montelukast, oral (US label, Bryant Ranch Prepack)

Full references p. 432–457